

Tuple

```
In [1]: tup1 = ()
```

```
In [2]: tup2 = (10,30,60)
tup2
```

```
Out[2]: (10, 30, 60)
```

```
In [3]: tup3 = (10.77,30.66,60.89)
```

```
In [4]: tup4 = ('one','two','three')
```

```
In [5]: tup5 = ('asif',25,(50,100),(150,90))
```

```
In [6]: tup6 =(100,'asif',17.765)
```

```
In [7]: tup7 = ('asif',25,[50,100],[150,90],{'john','david'},(99,22,33))
```

```
In [8]: tup7
```

```
Out[8]: ('asif', 25, [50, 100], [150, 90], {'david', 'john'}, (99, 22, 33))
```

Tuple Indexing

```
In [10]: tup2[0]
```

```
Out[10]: 10
```

```
In [11]: tup4[0]
```

```
Out[11]: 'one'
```

```
In [12]: tup4[0][0]
```

```
Out[12]: 'o'
```

```
In [13]: tup4[-1]
```

```
Out[13]: 'three'
```

```
In [14]: # tuple slicing
```

```
In [15]: tup7
```

```
Out[15]: ('asif', 25, [50, 100], [150, 90], {'david', 'john'}, (99, 22, 33))
```

```
In [16]: tup7[2:5]
```

```
Out[16]: ([50, 100], [150, 90], {'david', 'john'})
```

```
In [17]: tup7[-3:]
```

```
Out[17]: ([150, 90], {'david', 'john'}, (99, 22, 33))
```

```
In [18]: tup7[1]
```

```
Out[18]: 25
```

```
In [19]: tup7[:]
```

```
Out[19]: ('asif', 25, [50, 100], [150, 90], {'david', 'john'}, (99, 22, 33))
```

remove & change items

```
In [21]: tup7[4]
```

```
Out[21]: {'david', 'john'}
```

```
In [22]: tup7[4] = 101
```

```
-----  
TypeError                                Traceback (most recent call last)  
Cell In[22], line 1  
----> 1 tup7[4] = 101  
  
TypeError: 'tuple' object does not support item assignment
```

```
In [23]: for i in tup7:  
         print(i)
```

```
asif  
25  
[50, 100]  
[150, 90]  
{'john', 'david'}  
(99, 22, 33)
```

```
In [24]: for i in enumerate(tup7):  
         print(i)
```

```
(0, 'asif')  
(1, 25)  
(2, [50, 100])  
(3, [150, 90])  
(4, {'john', 'david'})  
(5, (99, 22, 33))
```

```
In [25]: 25 in tup7
```

```
Out[25]: True
```

index position

```
In [26]: tup7
```

```
Out[26]: ('asif', 25, [50, 100], [150, 90], {'david', 'john'}, (99, 22, 33))
```

```
In [29]: tup7.index('asif')
```

```
Out[29]: 0
```

sorting

```
In [31]: tup = (43,67,99,12,6,90,67)
```

```
In [32]: sorted(tup)
```

```
Out[32]: [6, 12, 43, 67, 67, 90, 99]
```

```
In [33]: sorted(tup,reverse =True)
```

```
Out[33]: [99, 90, 67, 67, 43, 12, 6]
```

```
In [ ]:
```